

Set B

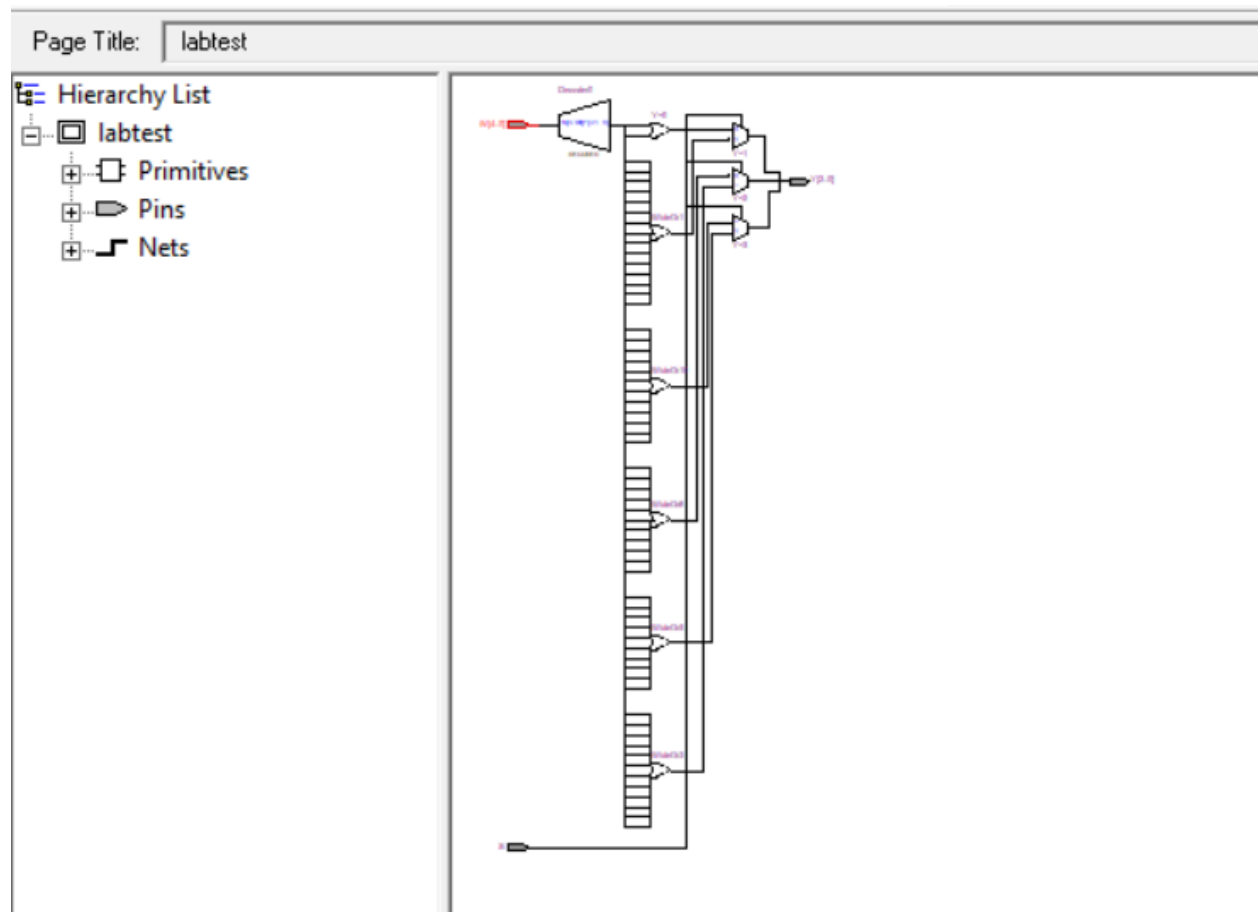
Answer to question no 2:

Code :

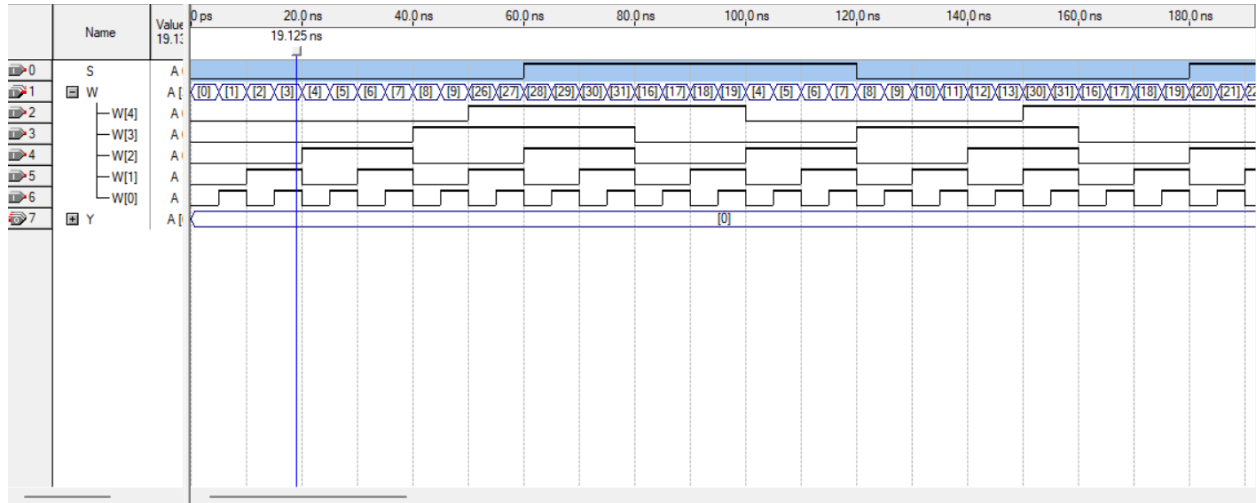
```
module labtest(W, S, Y);
    input [4:0] W;
    input S;
    output reg [2:0] Y;

    always @(W, S)
    begin
        if (S == 1)
        begin
            casex (W)
                5'b1xxxx: Y = 4;
                5'b01xxx: Y = 3;
                5'b001xx: Y = 2;
                5'b0001x: Y = 1;
                5'b00001: Y = 0;
                default: Y = 3'bx;
            endcase
        end
        else
        begin
            casex (W)
                5'bxx1xx: Y = 2;
                5'bxx01x: Y = 1;
                5'bx100x: Y = 3;
                5'b1000x: Y = 4;
                5'b00001: Y = 0;
                default: Y = 3'bx;
            endcase
        end
    end
endmodule
```

RTL:



Wave:



Answer to question no 1:

Code:

```
module labtest1(A,B,C,M,P,S,Z);
input A,B,C,M;
output P,Z;
output [0:1]S;

assign Z= P ^ M;
assign P= (~A & ~B & C)| ~C & (A ^ B);
assign S[0]= ~A & (B^C)| A & (~ (B^C));
assign S[1]= A&B&C | A & (B^C);

endmodule
```

Wave:



RTL Circuit:

